Subject Name: Web Programming **Register Number:**2403917710422119
Subject Code:22CS450

Ex.No: 3

Date: 14/02/26

Develop application to demonstrate applications of jQuery and Ajax.

Aim:
To develop a jQuery and Ajax based AI Mock Interview application where users can select a topic, fetch interview questions dynamically, submit their answers, and receive instant feedback.

Procedure:

1. Create a project folder with files: index.html, script.js, and server.js.
2. Design the HTML page with a topic selector dropdown, a question display area, an answer textarea,and a Submit Answer button.
3. Include the jQuery CDN in the HTML file.
4. Use $(document).ready() in script.js to initialize the application.
5. Use AJAX GET request to fetch a random interview question based on the selected topic.
6. Use AJAX POST request to submit the user's answer and receive AI feedback from the server.
7. Use AJAX DELETE request to clear/reset the current interview session.
8. Create a Node.js + Express backend with API routes: GET /question, POST /answer, DELETE /session.
9. The server stores question bank data in-memory and returns feedback based on keywords.
10. Run the server and open index.html in a browser to test the mock interview application.

Source Code: In file index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>AI Mock Interview - jQuery AJAX</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>  <style>      body { font-
family: Arial, sans-serif; max-width: 700px; margin: 40px auto; }     h2 { color: #2563eb; }
select, textarea, button { width: 100%; margin: 8px 0; padding: 10px; font-size: 14px; }
#feedback { background: #f0f9ff; border: 1px solid #bae6fd;                padding: 15px; border-
radius: 6px; margin-top: 10px; }     #question-box { background: #fefce8; border: 1px solid
#fde68a;                    padding: 15px; border-radius: 6px; font-weight: bold; }     .btn-primary {
background: #2563eb; color: white; border: none;                    border-radius: 6px; cursor:
pointer; }     .btn-danger  { background: #dc2626; color: white; border: none;
border-radius: 6px; cursor: pointer; }  </style>
</head>
<body>
  <h2>AI Mock Interview Platform</h2>

  <label>Select Topic:</label>
  <select id="topic">
    <option value="javascript">JavaScript</option>
    <option value="react">React</option>
    <option value="nodejs">Node.js</option>
<option value="database">Database</option>
  </select>
```

```
  <button class="btn-primary" id="getQuestion">Get Question</button>
  <hr>
  <div id="question-box">
    <p id="question-text">Click "Get Question" to start your interview.</p>
  </div>
  <br>  <textarea id="answer" rows="5"       placeholder="Type your
answer here..."></textarea>  <button class="btn-primary"
id="submitAnswer">Submit Answer</button>
   <button class="btn-danger"  id="resetSession">Reset Session</button>
  <div id="feedback" style="display:none;"></div>

  <script src="script.js"></script>
</body>
</html>
```

In file script.js

```
$(document).ready(function () {
let currentQuestionId = null;

  // Fetch a question via AJAX GET
$("#getQuestion").click(function () {     const
topic = $("#topic").val();     $.ajax({
      url: "http://localhost:3000/question",
method: "GET",       data: { topic: topic },
success: function (data) {          currentQuestionId
= data.id;         $("#question-
text").text(data.question);
        $("#answer").val("");
        $("#feedback").hide();       },       error: function () {
alert("Failed to fetch question. Is the server running?");       }
    });
  });

  // Submit answer via AJAX POST   $("#submitAnswer").click(function () {
const answer = $("#answer").val().trim();     if (!answer) { alert("Please
type your answer first."); return; }     if (!currentQuestionId) { alert("Fetch a
question first!"); return; }

    $.ajax({       url:
"http://localhost:3000/answer",
method: "POST",       contentType:
"application/json",        data:
JSON.stringify({          questionId:
currentQuestionId,          answer: answer
}),       success: function (data) {
$("#feedback")
          .html("<b>Feedback:</b><br>" + data.feedback +
                "<br><br><b>Score:</b> " + data.score + "/10")
          .show();
      }
    });
  });

  // Reset session via AJAX DELETE
  $("#resetSession").click(function () {     $.ajax({       url:
"http://localhost:3000/session",        method: "DELETE",       success:
```

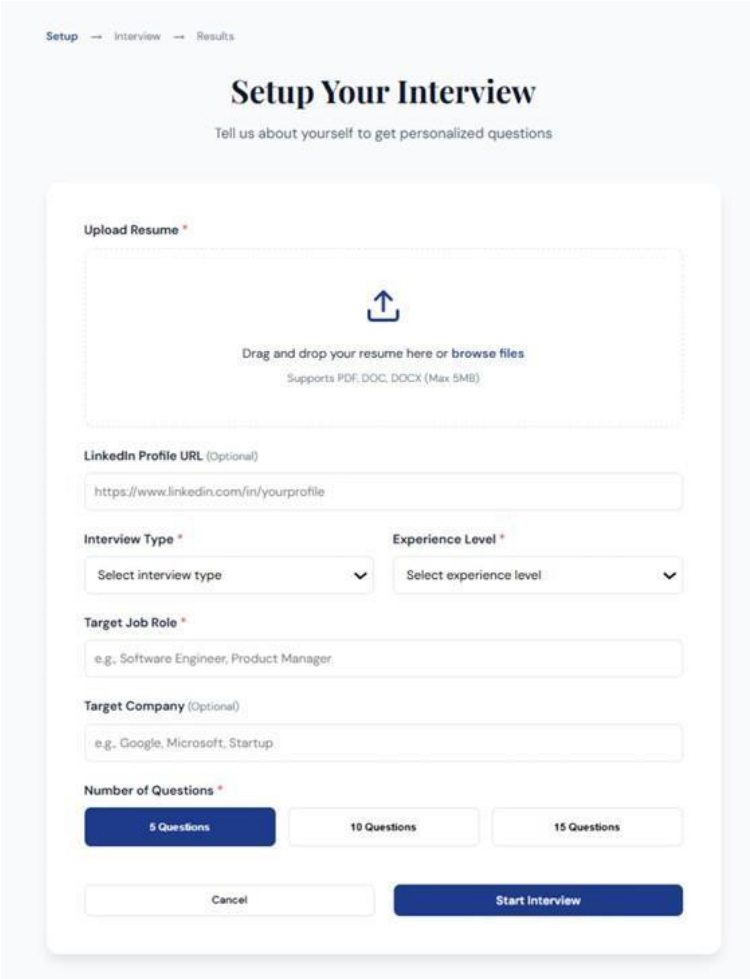
```
function () {        currentQuestionId = null;        $("#question-
text").text('Click "Get Question" to start.');
        $("#answer").val("");
$("#feedback").hide();        alert("Session
reset successfully!");       }
    });
  });
});
```

In file server.js

```
const express = require("express");
const cors   = require("cors"); const
app   =  express();  app.use(cors());
app.use(express.json());

const questions = {  javascript: [     { id: 1, question: "What is event
delegation in JavaScript?" },
    { id: 2, question: "Explain the difference between == and ===" }  ],
react: [     { id: 3, question: "What are React Hooks? Explain
useState." },
    { id: 4, question: "What is the Virtual DOM and how does it work?" }  ],
nodejs: [     { id: 5, question: "What is the event loop in Node.js?" },
    { id: 6, question: "Explain middleware in Express.js." }  ],   database: [
{ id: 7, question: "What is the difference between SQL and NoSQL?" },
    { id: 8, question: "Explain database indexing and its benefits." }
  ] };

const answerKeywords = {   1:
["bubbling","parent","child","delegation"],
  2: ["type","coercion","strict","equality"],
  3: ["state","functional","component","hook"],
  4: ["virtual","reconciliation","diff","render"],
  5: ["single","thread","non-blocking","callback"],
  6: ["request","response","next","pipeline"],
  7: ["schema","scalable","relational","document"],
  8: ["performance","query","search","column"]
};

// GET /question?topic=javascript app.get("/question", (req,
res) => {   const topic = req.query.topic || "javascript";   const
list  = questions[topic] || questions["javascript"];  const q       =
list[Math.floor(Math.random() * list.length)];   res.json(q); });

// POST /answer app.post("/answer", (req, res) => {   const { questionId, answer
} = req.body;   const keywords = answerKeywords[questionId] || [];   const lower
= answer.toLowerCase();   const matched  = keywords.filter(k =>
lower.includes(k)).length;  const score      = Math.min(10, Math.round((matched
/ keywords.length) * 10));   const feedback = score >= 7      ? "Excellent answer!
You covered the key concepts well."
    : score >= 4
    ? "Good attempt. Try to include more technical details."
: "Needs improvement. Review the topic and try again.";
res.json({ feedback, score }); });

// DELETE /session
app.delete("/session", (req, res) => {
res.send("Session cleared"); });
```

```
app.listen(3000, () =>
  console.log("Server running on http://localhost:3000"));
```

Output:



Result:

Thus the application to demonstrate the use of jQuery and Ajax in an AI Mock Interview platform is implemented successfully.

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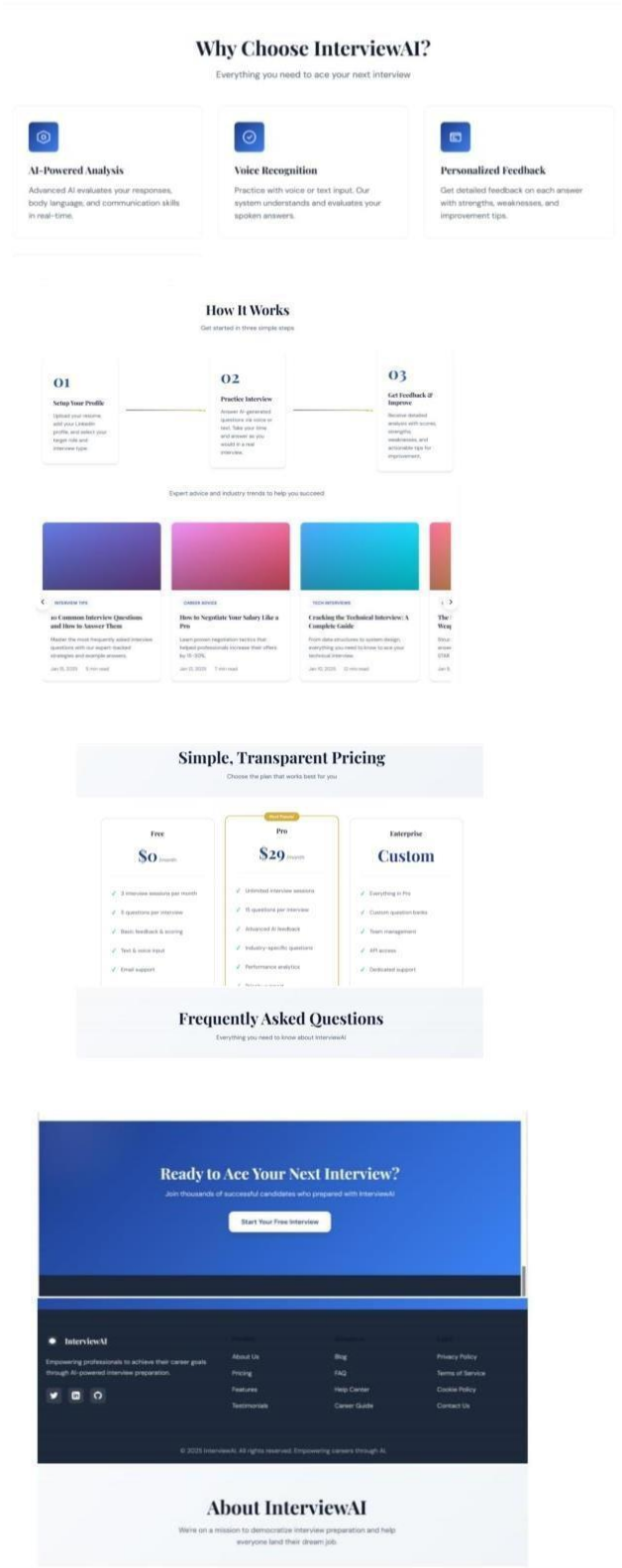
FULL STACK MINI PROJECT — AI MOCK INTERVIEW HUB

Folder Structure:

```
project/
│
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   ├── layout/
│   │   │   │   └── Navbar.jsx
│   │   │   │
│   │   │   ├── blog/
│   │   │   │   ├── BlogCard.jsx
│   │   │   │   ├── BlogList.jsx
│   │   │   │   └── BlogForm.jsx
│   │   │   │
│   │   │   └── auth/
│   │   │       ├── LoginForm.jsx
│   │   │       └── RegisterForm.jsx
│   │   │
│   │   ├── pages/
│   │   │   ├── Home.jsx
│   │   │   ├── Login.jsx
│   │   │   ├── Register.jsx
│   │   │   └── CreateBlog.jsx
│   │   │
│   │   ├── services/
│   │   │   └── api.js
│   │   │
│   │   ├── App.jsx
│   │   └── main.jsx
│   │
│   └── package.json

├── backend/
│   ├── config/
│   │   └── db.js
│   │
│   ├── models/
│   │   ├── User.js
│   │   └── Session.js
│   │
│   ├── controllers/
│   │   ├── userController.js
│   │   └── interviewController.js
│   │
│   ├── routes/
│   │   ├── userRoutes.js
│   │   └── interviewRoutes.js
│   │
│   ├── middleware/
│   │   └── authMiddleware.js
│   │
│   ├── src/
│   │   ├── components/
│   │   │   ├── layout/
│   │   │   │   └── Navbar.jsx
│   │   │   │
│   │   │   ├── interview/
│   │   │   │   ├── QuestionCard.jsx
│   │   │   │   ├── SessionList.jsx
│   │   │   │   └── InterviewForm.jsx
│   │   │   │
│   │   │   └── auth/
│   │   │       ├── LoginForm.jsx
│   │   │       └── RegisterForm.jsx
│   │   │
│   │   ├── pages/
│   │   │   ├── Home.jsx
│   │   │   ├── Login.jsx
│   │   │   ├── Register.jsx
│   │   │   └── Interview.jsx
│   │   │
│   │   ├── services/
│   │   │   └── api.js
│   │   │
│   │   ├── App.jsx
│   │   └── main.jsx
│   │
│   └── package.json
```

Output: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Practice with voice or text input. Our system understands and evaluates your spoken answers.
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02
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03
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Choose the plan that works best for you
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$0
Pro
$20
Enterprise
Custom
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